

YEAR 7 MATHEMATICS · CHAPTER 2

Angles

2.01 Naming angles

2.03 Classifying angles

2.05 Angles at a point and vertically opposite angles

2.07 Corresponding angles

2.09 Co-interior angles

2.11 Proving lines are parallel

2.02 Measuring and drawing angles

2.04 Complementary and supplementary angles

2.06 Parallel and perpendicular lines

2.08 Alternate angles

2.10 Finding unknown angles on parallel lines

Name _____

Class _____

Teacher _____

How to use this booklet

- Each lesson starts with **learning intentions** and **success criteria**. Write the **key terms** as your teacher explains them, and use the **notes** box.
- Work through the levels in order: Level 1, then Level 2, then Level 3. Extension questions stretch you further.
- Some questions ask you to **draw** a diagram, graph or construction. Use the space given, a ruler and a sharp pencil.
- At the end of each lesson, **show off your skill**: choose one question, then tear off the ticket and hand it in.
- Missed a lesson? Each lesson ends with a **summary** that has a worked example for every level. Start there, then try the questions.

WE DO

Grey boxes with a dark border. Work with your teacher. Copy what they write on the board.

YOU DO

Blue boxes. Work on your own. Show your working in the box and write your answer.

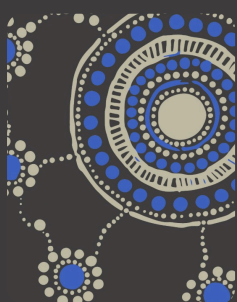
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In this chapter you will

- measure and draw angles with a protractor
- classify angles by size
- find complementary, supplementary and vertically opposite angles
- construct parallel and perpendicular lines
- find corresponding, alternate and co-interior angles on parallel lines
- prove that two lines are parallel

NSW Mathematics K–10 Syllabus (2022), Stage 4 Measurement and space: MA4-ANG-C-01 applies angle relationships to solve problems, including those related to transversals on sets of parallel lines.



Acknowledgement of artwork

The artwork used throughout this booklet series was created by **Marni Tuala**. We thank Marni for sharing this artwork with our school community.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
2.01

Naming angles

Date _____

Learning intentions

We are learning to:

- name an angle using three letters
- identify adjacent angles

Success criteria

I can:

- ☐ name an angle with the **vertex** in the middle
- ☐ draw an angle from its name
- ☐ name an **adjacent** angle

Key terms Write the definition as your teacher explains it.

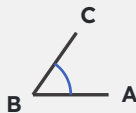
Vertex

Arm

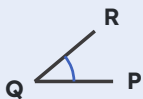
Adjacent angles

Notes

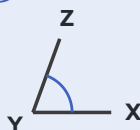
Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Name this angle in two ways.**Example 2** Draw and label $\angle BKT$.**YOU DO** On your own: Set A. Name each angle using three letters.

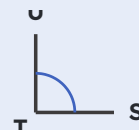
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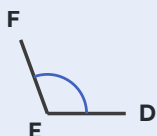
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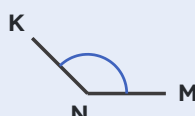
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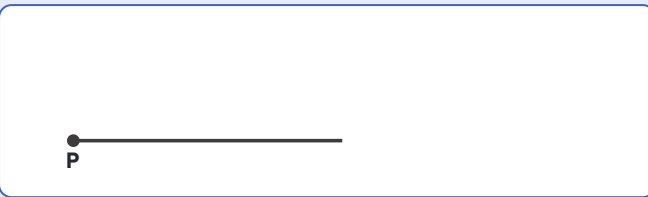
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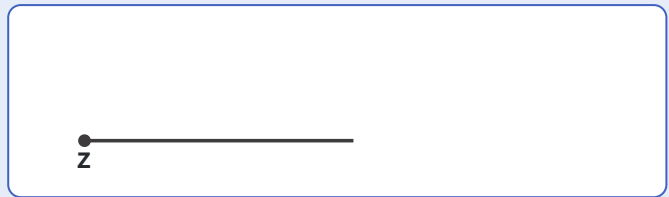
YOU DO

On your own: Set B. Draw and label each angle. Put the vertex at the dot.

1 $\angle FPR$



2 $\angle MZQ$



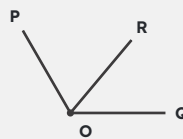
Level 2 · Building up

WE DO

With your teacher: copy what they write on the board.

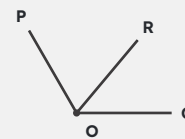
Example 3

Name the angle adjacent to $\angle POQ$.



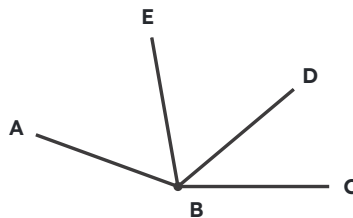
Example 4

Name all 3 angles at O.



YOU DO

On your own: Set C. Use the diagram.



1 Name the angle adjacent to $\angle ABE$ on the other side of BE.

4 How many angles are at B in total?

2 Name the angle adjacent to $\angle CBD$.

5 Which angle is made of $\angle DBE$ and $\angle EBA$?

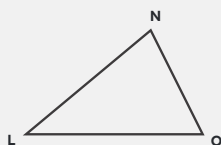
3 Name the largest angle in the diagram.

6 Name the angle between arms BC and BE.

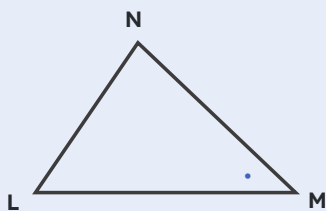
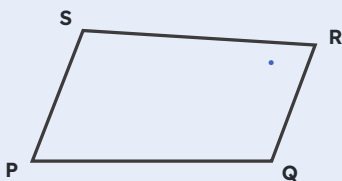


Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Name the angle marked with a •.

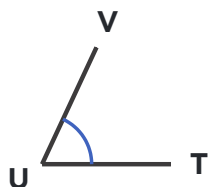
The • is at O.

Example 6 Why is it wrong to call an angle " $\angle B$ " when three lines meet at B?**YOU DO** On your own: Set D. Name the marked angle in each diagram.**1** The angle at vertex M in $\triangle LMN$.**3** Can $\angle ABC$ and $\angle CBA$ be the same angle? Explain.**2** The angle at R in quadrilateral PQRS.**4** Draw two adjacent angles $\angle PQR$ and $\angle RQS$.**YOU DO**

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

**Level 1**

Name the angle shown.

Level 2

Which letter is the vertex? Why?

Level 3Draw and label two adjacent angles $\angle ABC$ and $\angle CBD$.

Extension · Stretch yourself

Worked example

Four rays come from point O. How many different angles are there?

- Count angles using each pair of rays
- With 4 rays there are $3 + 2 + 1$ pairs

Answer: 6 angles

YOU DO

On your own: read the worked example, then try these.

- E1** How many angles are made by 5 rays from one point?

- E2** Name all the angles made by rays OA, OB, OC and OD.

Lesson summary

Missed the lesson? Start here.

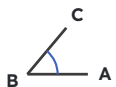
- Remember**
1. An angle has two **arms** and a **vertex**.
 3. $\angle ABC$ and $\angle CBA$ are the same angle.

2. Name it with 3 letters; the vertex goes in the middle.
4. **Adjacent** angles share a vertex and an arm.

Level 1

Name the angle

1. Vertex is B, arms go to A and C



Answer: $\angle ABC$

Level 2

Draw $\angle KLM$

1. Vertex L
2. Arms LK and LM

Answer: a V shape with L at the point

Level 3

Adjacent to $\angle POQ$?

1. Shares arm OQ and vertex O

Answer: $\angle QOR$

Extension

Angles from 4 rays

1. $3 + 2 + 1$

Answer: 6

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
2.02

Measuring and drawing angles

Date _____

Learning intentions

We are learning to:

- use a protractor to measure and draw angles, including reflex angles

Success criteria

I can:

- ☐ place the protractor centre on the vertex and 0° on one arm
- ☐ read the correct scale
- ☐ draw a **reflex** angle by drawing 360° minus the angle

Key terms Write the definition as your teacher explains it.

Protractor

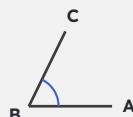
Reflex angle

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.

Example 1 Measure $\angle ABC$.



Example 2 Draw $\angle JUG = 84^\circ$.

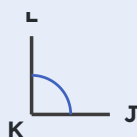


YOU DO On your own: Set A. Measure each angle.

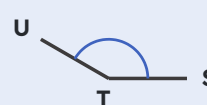
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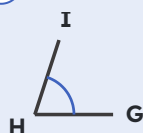
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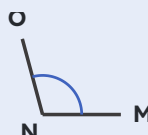
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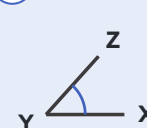
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4



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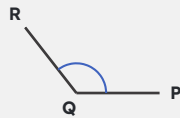


YOU DO On your own: Set B. Draw each angle.

1 $\angle POT = 41^\circ$



2 $\angle DGE = 150^\circ$

**Level 2 · Building up****WE DO** With your teacher: copy what they write on the board.**Example 3** Measure this obtuse angle.**Example 4** Draw $\angle QRA = 117^\circ$.**YOU DO** On your own: Set C. Draw each angle.

1 $\angle SAR = 96^\circ$



3 $\angle VWX = 165^\circ$



2 $\angle KLM = 23^\circ$



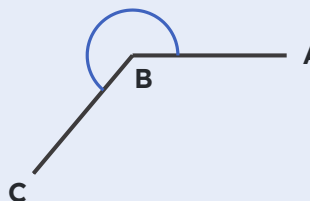
4 $\angle BCD = 72^\circ$



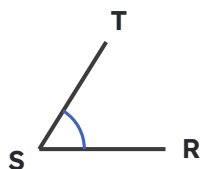


Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Draw $\angle XDW = 210^\circ$.**Example 6** Draw $\angle AMP = 300^\circ$.**YOU DO** On your own: Set D. Reflex angles.**1** Draw $\angle MNB = 195^\circ$ **3** Measure the reflex angle.**2** Draw $\angle PLO = 270^\circ$ **4** An angle is 145° . What size is the reflex angle at the same vertex?**YOU DO** Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

**Level 1**Measure $\angle RST$.**Level 2**Draw $\angle ABC = 135^\circ$.**Level 3**Draw $\angle XYZ = 250^\circ$.

Extension · Stretch yourself

Worked example

The hands of a clock show 4 o'clock. What is the angle between them? What is the reflex angle?

1. Each hour mark is $360 \div 12 = 30^\circ$
2. 4 hours apart: $4 \times 30 = 120^\circ$
3. Reflex: $360 - 120 = 240^\circ$

Answer: 120° and reflex 240°

YOU DO

On your own: read the worked example, then try these.

- E1** Find the angle between the hands at 5 o'clock and at 9 o'clock.

- E2** At 3:30 the hour hand is halfway between 3 and 4. Find the angle between the hands.

Lesson summary

Missed the lesson? Start here.

Remember

1. Centre of the protractor on the vertex.
2. 0° line along one arm.
3. Count up from 0° on that scale.
4. Reflex: measure the small angle, then $360^\circ - \text{it}$.

Level 1

Measure a 64° angle

1. Centre on vertex, 0° on arm
2. Read the scale that starts at 0°

Answer: 64°

Level 2

Draw 117°

1. Draw an arm, mark 117° , join to the vertex

Answer: an obtuse angle

Level 3

Draw 210°

1. $360 - 210 = 150$
2. Draw 150° the other way

Answer: a reflex angle

Extension

Angle at 4 o'clock

1. $4 \times 30^\circ$

Answer: 120°

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
2.03

Classifying angles

Date _____

Learning intentions

We are learning to:

- classify angles as acute, right, obtuse, straight, reflex or a revolution

Success criteria

I can:

- ☐ know the size range for each type of angle
- ☐ classify an angle from its size or a diagram
- ☐ name and classify an angle in a diagram

Key terms Write the definition as your teacher explains it.

Acute

Obtuse

Reflex

Notes

Level 1 · Getting started

WE DO

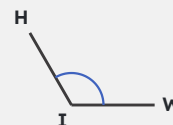
With your teacher: copy what they write on the board.

Example 1

Classify: 34° , 90° , 145° , 180° , 250° , 360°

Example 2

Name and classify the angle.



YOU DO

On your own: Set A. Classify each angle.

1 47° 4 180° 2 90° 5 215° 3 160° 6 12°

YOU DO On your own: Set B. Classify each angle.

1 89°

4 360°

2 91°

5 179°

3 359°

6 181°

Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

Example 3 Is 90.5° acute, right or obtuse?

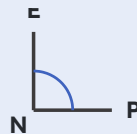
Example 4 An angle is 3 times a right angle. Classify it.

YOU DO On your own: Set C. Name each angle with 3 letters, then classify it.

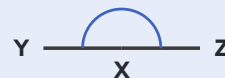
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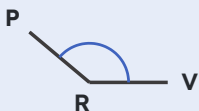
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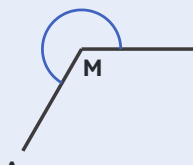
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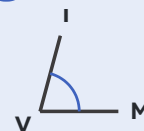
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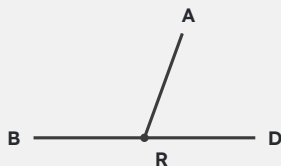
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Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Name and classify both angles at R.**Example 6** Can a triangle have an obtuse angle and a right angle? Explain.**YOU DO** On your own: Set D. Think it through.

- 1 What angle does a minute hand turn through in 20 minutes? Classify it.

- 3 List 3 obtuse angles that are multiples of 25° .

- 2 Two acute angles are added. Can the answer be obtuse? Give an example.

- 4 How many reflex angles are there between 200° and 220° (whole degrees)?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1Classify 88° .**Level 2**Classify 200° and 180° .**Level 3**Name and classify both angles in Example 5 ($\angle DRA$ and $\angle ARB$).

Extension · Stretch yourself

Worked example

A wheel turns $1\frac{3}{4}$ revolutions. How many degrees is that?

1. 1 revolution = 360°
2. $\frac{3}{4}$ of $360^\circ = 270^\circ$
3. $360 + 270$

Answer: 630°

YOU DO

On your own: read the worked example, then try these.

- E1** A skater spins $2\frac{1}{2}$ times. How many degrees?

- E2** A fan turns 1080° . How many revolutions?

Lesson summary

Missed the lesson? Start here.

- Angle types**
1. **Acute:** less than 90° . **Right:** 90° .
 2. **Obtuse:** between 90° and 180° .
 3. **Straight:** 180° . **Reflex:** between 180° and 360° .
 4. **Revolution:** 360° .

Level 1

Classify 125°

1. Between 90° and 180°

Answer: obtuse

Level 2

Classify 300°

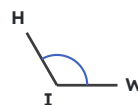
1. Between 180° and 360°

Answer: reflex

Level 3

Name and classify

1. Vertex I in the middle
2. More than 90°



Answer: $\angle WIH$, obtuse

Extension

$1\frac{3}{4}$ revolutions

1. $360 + 270$

Answer: 630°

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LESSON
2.04

Complementary and supplementary angles

Date _____

Learning intentions

We are learning to:

- find complementary and supplementary angles

Success criteria

I can:

- ☐ know complementary angles add to 90°
- ☐ know supplementary angles add to 180°
- ☐ find an unknown angle in a right angle or on a straight line

Key terms Write the definition as your teacher explains it.**Complementary****Supplementary**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Find the complement of 35° and of 78° .**Example 2** Find the supplement of 45° and of 100° .**YOU DO** On your own: Set A. Write the complement of each angle.1 35° 2 78° 3 4° 4 60° 5 45° 6 89°

YOU DO On your own: Set B. Write the supplement of each angle.

1 45°

4 90°

2 100°

5 12°

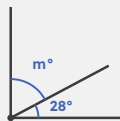
3 178°

6 135°

Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

Example 3 Find the value of m .

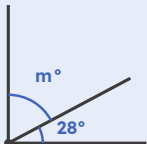


Example 4 Find the value of k .

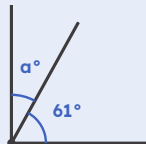


YOU DO On your own: Set C. Find the value of each variable.

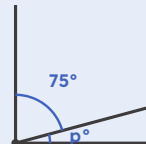
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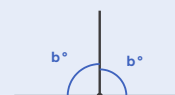
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4



6





Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Find x .**Example 6** Two supplementary angles are equal. What size are they?**YOU DO** On your own: Set D. Harder problems.

- 1 An angle is 20° more than its complement. Find both angles.

- 3 Find x if $3x^\circ$ and 60° are supplementary.

- 2 The supplement of an angle is 4 times the angle. Find the angle.

- 4 Can an obtuse angle have a complement? Explain.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1Find the complement of 62° .**Level 2**Find the supplement of 118° .**Level 3**Two angles on a straight line are x° and $3x^\circ$. Find x .

Extension · Stretch yourself

Worked example

Find the angle whose supplement is 3 times its complement.

1. Let the angle be x
2. $180 - x = 3(90 - x)$
3. $180 - x = 270 - 3x$, so $2x = 90$

Answer: $x = 45^\circ$

YOU DO

On your own: read the worked example, then try these.

- E1** Find the angle whose supplement is 4 times its complement.

- E2** Find the complement of the supplement of 130° .

Lesson summary

Missed the lesson? Start here.

Remember

1. **Complementary** angles add to 90° .
2. **Supplementary** angles add to 180° .
3. Right angle split in two → complementary.
4. Straight line split in two → supplementary.

Level 1

Complement of 35°

1. $90 - 35$

Answer: 55°

Level 2

Supplement of 100°

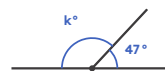
1. $180 - 100$

Answer: 80°

Level 3

Find k

1. $k + 47 = 180$



Answer: $k = 133$

Extension

Supplement = $3 \times$ complement

1. $180 - x = 270 - 3x$

Answer: 45°

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LESSON
2.05

Angles at a point and vertically opposite angles

Date _____

Learning intentions

We are learning to:

- find unknown angles using angle rules

Success criteria

I can:

- ☐ use angles on a straight line add to 180°
- ☐ use angles at a point add to 360°
- ☐ use vertically opposite angles are equal

Key terms Write the definition as your teacher explains it.

Angles at a point

Vertically opposite

Notes

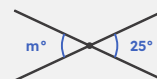
Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.

Example 1 Find a .



Example 2 Find m .



YOU DO On your own: Set A. Angles on a straight line. Find each variable.

1



3



5



2



4

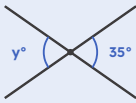


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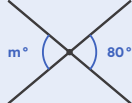


YOU DO On your own: Set B. Vertically opposite angles. Find each variable.

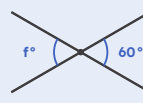
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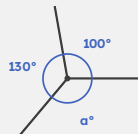
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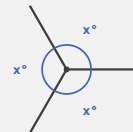
Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

Example 3 Find a (angles at a point).

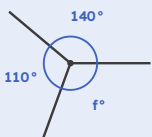


Example 4 Find x .

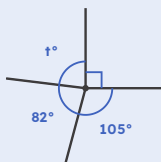


YOU DO On your own: Set C. Angles at a point. Find each variable.

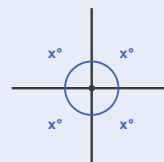
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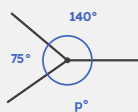


Level 3 · Challenge

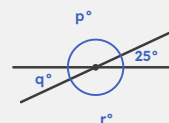
Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.

Example 5 Find p .

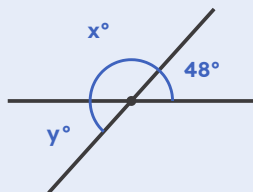


Example 6 Find p , q and r .



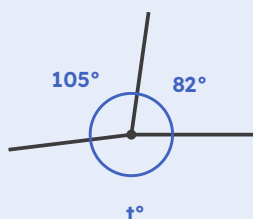
YOU DO On your own: Set D. Find each variable. Give a reason.

1 Find x and y .



3 Three equal angles meet at a point. Find each angle.

2 Find t .



4 Two lines cross. One angle is 65° . Find the other three angles.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Angles on a straight line: 64° and x° . Find x .

Level 2

Two lines cross. One angle is 38° . Find the vertically opposite angle.

Level 3

Angles at a point: 95° , 130° and y° . Find y .

Extension · Stretch yourself

Worked example

Angles at a point are $2x^\circ$, $3x^\circ$ and $4x^\circ$. Find x .

- Angles at a point add to 360°
- $2x + 3x + 4x = 9x = 360$

Answer: $x = 40$

YOU DO

On your own: read the worked example, then try these.

- E1** Angles on a straight line are x° , $2x^\circ$ and $3x^\circ$. Find each angle.

- E2** Two lines cross. One angle is $(2x + 10)^\circ$ and its vertically opposite angle is 70° . Find x .

Lesson summary

Missed the lesson? Start here.

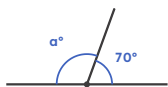
- Remember**
1. Straight line: angles add to 180° .
 3. **Vertically opposite** angles are equal.

2. At a point: angles add to 360° .
4. Always give a reason.

Level 1

Find a

- $a + 70 = 180$

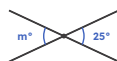


Answer: $a = 110$

Level 2

Find m

- Vertically opposite



Answer: $m = 25$

Level 3

At a point: 140° , 75° , p°

- $p = 360 - 215$

Answer: $p = 145$

Extension

$$2x + 3x + 4x = 360$$

- $9x = 360$

Answer: $x = 40$

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LESSON
2.06

Parallel and perpendicular lines

Date _____

Learning intentions

We are learning to:

- identify parallel and perpendicular lines
- construct parallel and perpendicular lines with compasses

Success criteria

I can:

- ☐ use the symbols $\parallel$ and $\perp$
- ☐ construct a line through a point parallel to another line
- ☐ construct a line through a point perpendicular to another line

Key terms Write the definition as your teacher explains it.

Parallel

Perpendicular

Intersect

Notes

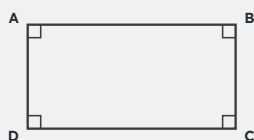
Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

Name 2 parallel lines and 2 perpendicular lines.



Example 2

What does $AB \perp CD$ mean? What does $EF \parallel GH$ mean?

YOU DO

On your own: Set A. Use the rectangle ABCD in Example 1.

- 1 Name a line parallel to AB.

- 4 Is $AC \perp BD$? (Draw the diagonals.)

- 2 Name a line perpendicular to AB.

- 5 Where do AB and BC intersect?

- 3 Name a line parallel to BC.

- 6 Write AD and BC using a symbol.

YOU DO On your own: Set B. True or false.

- | | | | |
|--|----------------------|---|----------------------|
| 1 Parallel lines never meet. | <input type="text"/> | 3 Two lines that intersect must be perpendicular. | <input type="text"/> |
| 2 Perpendicular lines meet at 90° . | <input type="text"/> | 4 Railway tracks are parallel. | <input type="text"/> |

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Construct a line through P parallel to BC.**Example 4** Construct a line through X perpendicular to BC (X on BC).**YOU DO** On your own: Set C. Constructions. Use a ruler and compasses.

- | | |
|---|--|
| 1 Draw a line BC and point P above it. Construct a line through P parallel to BC. | 2 Draw a line BC with X on it. Construct a line through X perpendicular to BC. |
|---|--|



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Construct a line through X perpendicular to BC (X not on BC).

Example 6 Name everyday examples of parallel and perpendicular lines.

YOU DO On your own: Set D. Constructions.**1** Draw a line BC and point X below it. Construct the perpendicular from X to BC.

2 Construct a square with side 4 cm using perpendicular lines.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

What symbol means “is parallel to”?

Level 2

Name 2 perpendicular lines in your classroom.

Level 3

Construct a line through a point P perpendicular to a line (draw your own).

Extension · Stretch yourself

Worked example

Construct the perpendicular bisector of a 6 cm interval AB.

1. Set compasses to more than half of AB
2. Draw arcs above and below AB from A and from B
3. Join the two crossing points

Answer: a line perpendicular to AB through its midpoint

YOU DO

On your own: read the worked example, then try these.

- E1** Draw a 7 cm interval and construct its perpendicular bisector.

- E2** Why does the construction work? (Hint: equal distances.)

Lesson summary

Missed the lesson? Start here.

Remember 1. **Parallel** ($\parallel$): same direction, never meet.
3. Lines that cross **intersect**.

2. **Perpendicular** ($\perp$): meet at 90° .
4. Constructions use compass arcs of equal radius.

Level 1

Rectangle ABCD

1. Opposite sides

Answer: $AB \parallel DC$, $AB \perp BC$

Level 2

Parallel through P

1. Copy the angle or use two equal arcs

Answer: a line through P that never meets BC

Level 3

Perpendicular through X on BC

1. Equal arcs either side of X
2. Arcs above from those points

Answer: join X to the crossing point

Extension

Perpendicular bisector

1. Arcs from both ends

Answer: join the crossings

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
2.07

Corresponding angles

Date _____

Learning intentions

We are learning to:

- identify corresponding angles on parallel lines

Success criteria

I can:

- ☐ find **corresponding** angles (F shape)
- ☐ know they are equal on parallel lines
- ☐ find unknown angles using them

Key terms Write the definition as your teacher explains it.

Transversal

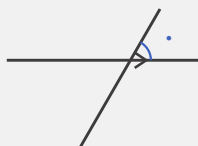
Corresponding angles

Notes

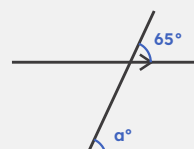
Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.

Example 1 Mark the angle corresponding to the one shown.



Example 2 Find a .



YOU DO On your own: Set A. Mark the corresponding angle on the diagram.

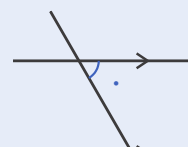
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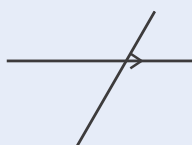
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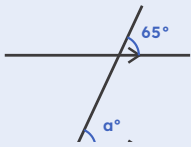


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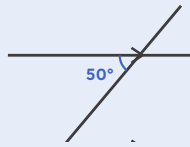


YOU DO On your own: Set B. Find each variable.

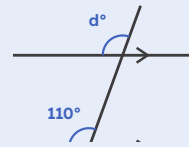
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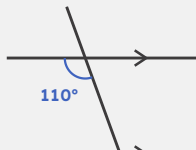
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Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

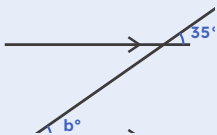
Example 3 Find m .



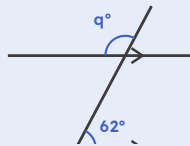
Example 4 Where is the F shape in corresponding angles?

YOU DO On your own: Set C. Find each variable. Give a reason.

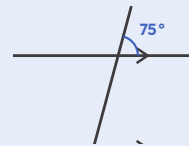
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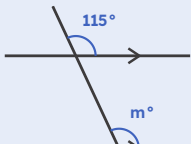
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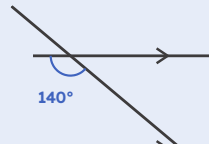
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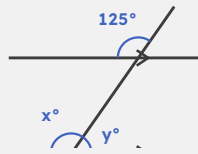
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Level 3 · Challenge

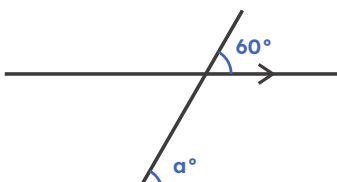
Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Find x and y .**Example 6** Are corresponding angles always equal? Explain.**YOU DO** On your own: Set D. Think it through.**1** Explain what a transversal is.
3 If the lines are not parallel, are corresponding angles equal?
2 Draw two parallel lines and a transversal. Mark a pair of corresponding angles.
4 In Set C question 3, which angle rule did you use besides corresponding angles?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

**Level 1**Find a .
Level 2

What shape do corresponding angles make?

Level 3Corresponding angles are $(2x + 20)^\circ$ and 60° . Find x .

Extension · Stretch yourself

Worked example

Corresponding angles are $(3x + 5)^\circ$ and 50° . Find x .

- Corresponding angles on parallel lines are equal
- $3x + 5 = 50$
- $3x = 45$

Answer: $x = 15$

YOU DO

On your own: read the worked example, then try these.

- E1 Corresponding angles are $(2y - 10)^\circ$ and 70° . Find y .

- E2 Corresponding angles are $4a^\circ$ and $(a + 90)^\circ$. Find a .

Lesson summary

Missed the lesson? Start here.

- Remember**
- A transversal crosses two or more lines.
 - On parallel lines they are **equal**.

- Corresponding angles are in matching positions (F shape).
- Reason: "corresponding angles, $AB \parallel CD$ ".

Level 1

Corresponding to top-right?

- Same position at the other line

Answer: bottom-right

Level 2

Find a

- Corresponding angles



Answer: $a = 65$

Level 3

Find x then y

- $x = 125$ corresponding
- $y = 180 - 125$

Answer: $x = 125, y = 55$

Extension

$$3x + 5 = 50$$

- $3x = 45$

Answer: $x = 15$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
2.08

Alternate angles

Date _____

Learning intentions

We are learning to:

- identify alternate angles on parallel lines

Success criteria

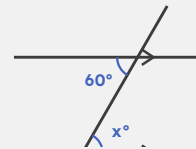
I can:

- ☐ find **alternate** angles (Z shape)
- ☐ know they are equal on parallel lines
- ☐ find unknown angles using them

Key terms Write the definition as your teacher explains it.**Alternate angles****Z shape**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Mark the angle alternate to the one shown.**Example 2** Find x .**YOU DO** On your own: Set A. Mark the alternate angle on the diagram.

1



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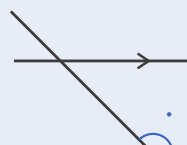
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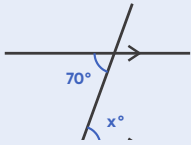


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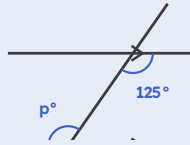


YOU DO On your own: Set B. Find each variable.

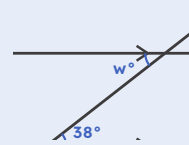
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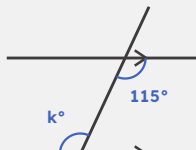
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Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

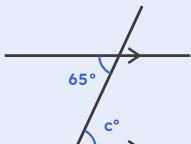
Example 3 Find k .



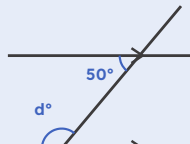
Example 4 Alternate angles are between the parallel lines. True or false?

YOU DO On your own: Set C. Find each variable. Give a reason.

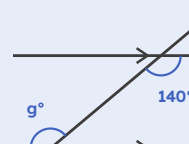
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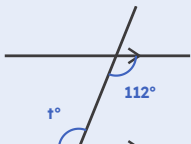
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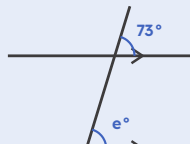
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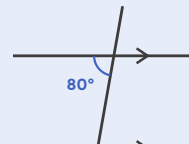
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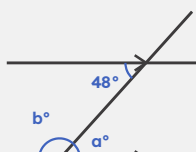


Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.

Example 5 Find a and b .



Example 6 Why is it called a Z shape?

YOU DO On your own: Set D. Think it through.

1 Draw a Z shape and mark a pair of alternate angles.

3 In Set C question 3, what two rules did you use?

2 Alternate angles are $(x + 30)^\circ$ and 75° . Find x .

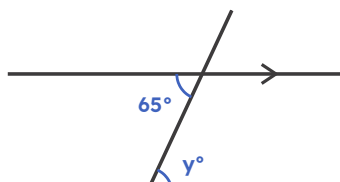
4 Two alternate angles are $2y^\circ$ and 90° . What can you say about the transversal?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____



Level 1

Find y .

Level 2

What letter shape do alternate angles make?

Level 3

Alternate angles are $(3x - 15)^\circ$ and 90° . Find x .

Extension · Stretch yourself

Worked example

Alternate angles are $5x^\circ$ and $(2x + 45)^\circ$. Find x and the angle size.

1. Alternate angles are equal
2. $5x = 2x + 45$, so $3x = 45$
3. $x = 15$, angle = 5×15

Answer: $x = 15$, angle 75°

YOU DO

On your own: read the worked example, then try these.

- E1** Alternate angles are $(4a - 20)^\circ$ and $(2a + 30)^\circ$. Find a .

- E2** Alternate angles are $3m^\circ$ and $(m + 70)^\circ$. Find both angles.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. **Alternate** angles are on opposite sides of the transversal.
 2. They are between the parallel lines (Z shape).
 3. On parallel lines they are **equal**.
 4. Reason: "alternate angles, $AB \parallel CD$ ".

Level 1

Alternate to top lower-left?

1. Other side, other line, inside

Answer: bottom upper-right

Level 2

Find x

1. Alternate angles



Answer: $x = 60$

Level 3

Find a and b

1. $a = 48$ alternate
2. $b = 180 - 48$

Answer: $a = 48$, $b = 132$

Extension

$$5x = 2x + 45$$

1. $3x = 45$

Answer: $x = 15$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
2.09

Co-interior angles

Date _____

Learning intentions

We are learning to:

- identify co-interior angles on parallel lines

Success criteria

I can:

- ☐ find **co-interior** angles (C shape)
- ☐ know they add to 180° on parallel lines
- ☐ find unknown angles using them

Key terms Write the definition as your teacher explains it.

Co-interior angles

Supplementary

Notes

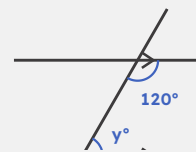
Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.

Example 1 Mark the angle co-interior to the one shown.



Example 2 Find y .



YOU DO On your own: Set A. Mark the co-interior angle on the diagram.

1



3



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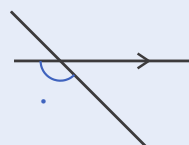
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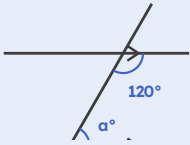


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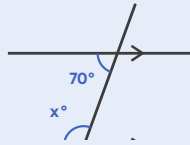


YOU DO On your own: Set B. Find each variable.

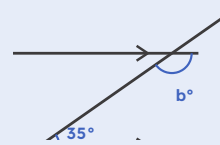
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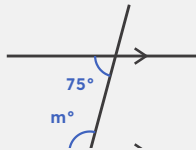
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Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

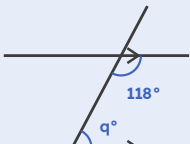
Example 3 Find m .



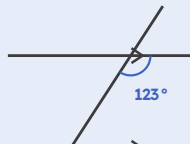
Example 4 Are co-interior angles equal? Explain.

YOU DO On your own: Set C. Find each variable. Give a reason.

1



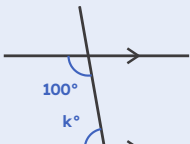
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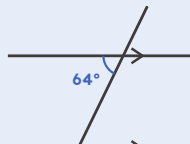
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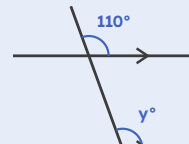
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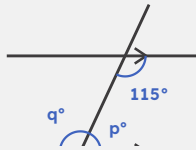
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Level 3 · Challenge

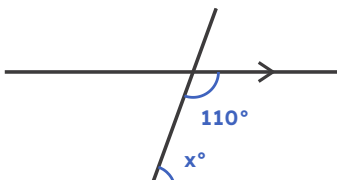
Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Find p and q .**Example 6** Name the three angle pairs on parallel lines and their rules.**YOU DO** On your own: Set D. Think it through.**1** Co-interior angles are x° and $2x^\circ$. Find x .**3** Draw a C shape and mark co-interior angles.**2** Co-interior angles are equal. What size are they? What does this tell you about the transversal?**4** Co-interior angles are $(x + 40)^\circ$ and $(x + 20)^\circ$. Find x .

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

**Level 1**Find x .**Level 2**

What do co-interior angles add to on parallel lines?

Level 3Co-interior angles are $3y^\circ$ and $6y^\circ$. Find y .

Extension · Stretch yourself

Worked example

Co-interior angles are $(2x + 10)^\circ$ and $(3x - 30)^\circ$. Find x .

1. Co-interior angles add to 180°
2. $2x + 10 + 3x - 30 = 180$
3. $5x - 20 = 180$, $5x = 200$

Answer: $x = 40$

YOU DO

On your own: read the worked example, then try these.

- E1** Co-interior angles are $4y^\circ$ and $5y^\circ$. Find both angles.

- E2** Co-interior angles are $(a + 25)^\circ$ and $(2a + 5)^\circ$. Find a .

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Co-interior angles are on the same side of the transversal.
 2. They are between the parallel lines (C shape).
 3. On parallel lines they add to 180° .
 4. Reason: "co-interior angles, $AB \parallel CD$ ".

Level 1

Co-interior to top lower-right?

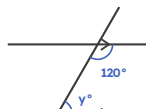
1. Same side, inside

Answer: bottom upper-right

Level 2

Find y

1. $y + 120 = 180$



Answer: $y = 60$

Level 3

p and q

1. $p = 180 - 115 = 65$
2. $q = 180 - 65$

Answer: $p = 65$, $q = 115$

Extension

$$(2x + 10) + (3x - 30) = 180$$

1. $5x = 200$

Answer: $x = 40$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
2.10

Finding unknown angles on parallel lines

Date _____

Learning intentions

We are learning to:

- use angle rules to find unknown angles on parallel lines

Success criteria

I can:

- ☐ choose the right parallel-line rule
- ☐ combine it with other angle rules
- ☐ give a reason for every step

Key terms Write the definition as your teacher explains it.

Reason

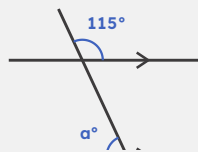
Parallel lines

Notes

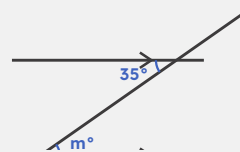
Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.

Example 1 Find a . Give a reason.

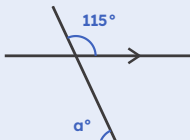


Example 2 Find m . Give a reason.

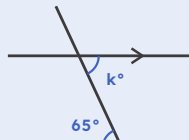


YOU DO On your own: Set A. Find each variable.

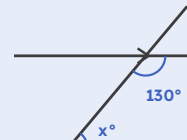
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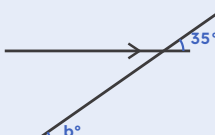
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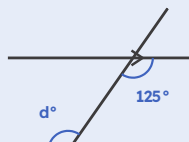
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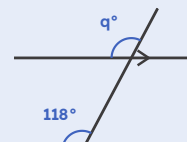
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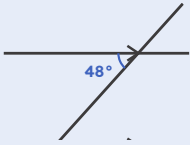


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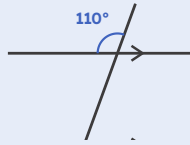


YOU DO On your own: Set B. Find each variable.

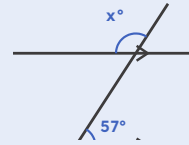
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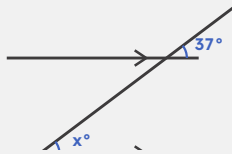
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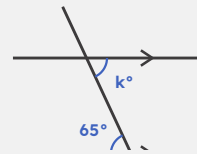
Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

Example 3 Find x and y .

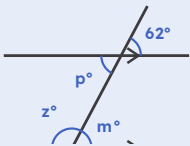


Example 4 Find k .

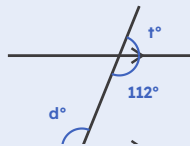


YOU DO On your own: Set C. Find all the variables. Give reasons.

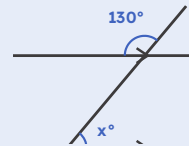
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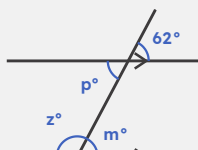
3





Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Find p , m and z .**Example 6** Explain how you would find an angle if no rule gives it directly.**YOU DO** On your own: Set D. Harder diagrams.

- 1 Two parallel lines are cut by a transversal. One angle is 57° . Find all 8 angles.

- 3 Explain why the 4 angles at the top intersection match the 4 at the bottom.

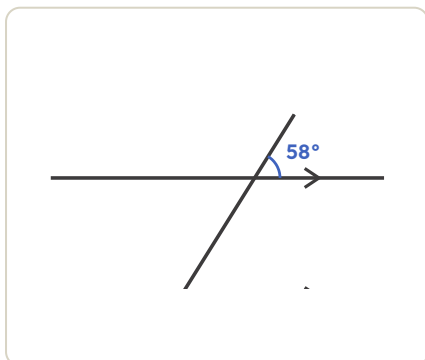
- 2 An angle on parallel lines is 3 times its co-interior angle. Find both angles.

- 4 Write your own question using alternate angles, with the answer 72° .

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

**Level 1**Find x .

Level 2

Give the reason you used.

Level 3Find all 8 angles when one angle is 58° .

Extension · Stretch yourself

Worked example

Find x if one angle is $(5x - 10)^\circ$ and its co-interior angle is $(3x + 30)^\circ$.

1. Co-interior angles add to 180°
2. $8x + 20 = 180$
3. $8x = 160$

Answer: $x = 20$ (angles 90° and 90°)

YOU DO

On your own: read the worked example, then try these.

- E1** Alternate angles are $(7x - 12)^\circ$ and $(4x + 36)^\circ$. Find x .

- E2** Corresponding angles are $(x + 15)^\circ$ and $(2x - 25)^\circ$. Find x .

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Corresponding and alternate angles are **equal**.
 2. Co-interior angles add to **180°** .
 3. Use straight lines (180°) and vertically opposite angles too.
 4. Give a **reason** for each step.

Level 1

Find a

1. Top ur $115^\circ \rightarrow$ bottom ur 115° (corr.)
2. $a = 180 - 115$ (straight line)

Answer: $a = 65$

Level 2

Find m

1. Alternate angles

Answer: $m = 35$

Level 3

p, m, z

1. $p = 62$ (vert. opp.)
2. $m = 62$ (corr.)
3. $z = 118$ (straight line)

Answer: $p = 62, m = 62, z = 118$

Extension

$$5x - 10 + 3x + 30 = 180$$

1. $8x = 160$

Answer: $x = 20$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
2.11

Proving lines are parallel

Date _____

Learning intentions

We are learning to:

- use angle rules to prove that two lines are parallel

Success criteria

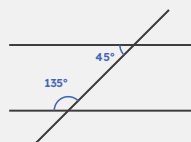
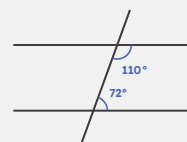
I can:

- ☐ check whether corresponding or alternate angles are equal
- ☐ check whether co-interior angles add to 180°
- ☐ write a conclusion: $AB \parallel CD$ with a reason

Key terms Write the definition as your teacher explains it.**Prove****Converse**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Is $AB \parallel CD$? Give a reason.**Example 2** Is $AB \parallel CD$? Give a reason.**YOU DO** On your own: Set A. Is $AB \parallel CD$? Write yes or no.

- 1 Corresponding angles:
 70° and 70°

- 4 Co-interior angles: 100°
and 100°

- 2 Alternate angles: 55°
and 65°

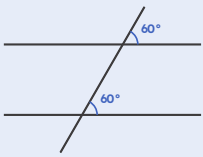
- 5 Alternate angles: 38°
and 38°

- 3 Co-interior angles: 120°
and 60°

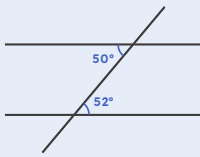
- 6 Corresponding angles:
 91° and 89°

YOU DO On your own: Set B. Is $AB \parallel CD$? Give a reason.

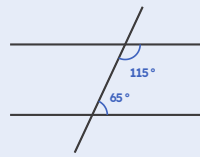
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2



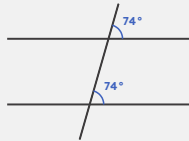
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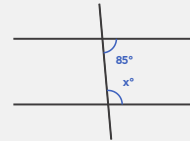
Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

Example 3 Is $AB \parallel CD$? Give a reason.



Example 4 Which value of x proves $AB \parallel CD$?



YOU DO On your own: Set C. Find the value of x that makes $AB \parallel CD$.

1 Corresponding: 112° and x°

3 Co-interior: 95° and x°

5 Alternate: $(x + 10)^\circ$ and 80°

2 Alternate: 47° and x°

4 Co-interior: $2x^\circ$ and 70°

6 Corresponding: $3x^\circ$ and 111°



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** What are the three tests for parallel lines?

Example 6 Why do we write “converse”?

YOU DO On your own: Set D. Write a proof.

- 1 A transversal cuts two lines. Alternate angles are 64° and 64° . Write a full proof that the lines are parallel.

- 3 Corresponding angles are 45° and 54° . Are the lines parallel? Explain.

- 2 Co-interior angles are 108° and 72° . Prove the lines are parallel.

- 4 Draw a diagram where the lines are NOT parallel and explain how you know.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1Alternate angles are 62° and 62° . Are the lines parallel?

Level 2Co-interior angles are 105° and 85° . Are the lines parallel? Explain.

Level 3Find x so that corresponding angles $(2x + 6)^\circ$ and 80° make $AB \parallel CD$.

Extension · Stretch yourself

Worked example

A line crosses AB and CD. The vertically opposite angle to an angle on AB is 72° , and the alternate angle on CD is 72° . Is $AB \parallel CD$?

- Vertically opposite angles are equal, so the angle on AB is 72°
- Alternate angles 72° and 72° are equal

Answer: Yes, $AB \parallel CD$ (alternate angles equal)

YOU DO

On your own: read the worked example, then try these.

- E1** Angles on a straight line on AB are 48° and x° . The co-interior angle to x° on CD is 48° . Is $AB \parallel CD$?

- E2** Why is it not enough to measure the angles with a protractor to prove lines are parallel?

Lesson summary

Missed the lesson? Start here.

Tests for parallel lines

- Corresponding angles are **equal**, or
- alternate angles are **equal**, or
- co-interior angles add to **180°** .
- Write: " $AB \parallel CD$ (alternate angles are equal)".

Level 1

135° and 45° co-interior?

- $135 + 45 = 180$

Answer: $AB \parallel CD$

Level 2

110° and 72° co-interior?

- $110 + 72 = 182 \neq 180$

Answer: not parallel

Level 3

74° and 74° corresponding

- Equal

Answer: $AB \parallel CD$

Extension

Co-interior 85° and x°

- $x = 180 - 85$

Answer: $x = 95$

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